Lecture 15 // Oct. 13 // Cobordism Maps

Last time: Given a cobordism $W$ from $Y_1$ to $Y_2$, we defined

$$M_{[\mathfrak{s}]}(x, W^*, y) = \left\{ \begin{array}{l} \text{ASD conn. on } W^* \\ \text{limiting to } x, y \\ \text{in component } \mathfrak{s} \end{array} \right\} / \text{gauge}$$

Note that there are $\mathbb{Z}$-many components.

Today, we'll continue defining the chain maps induced by $W$.

We'll restrict cobordisms $W$ to those which satisfy $b_2^+(W) = b_1(W) = 0$.

Let's define a grading-preserving chain map $F_W$ in the $C$, $\bar{C}$, and $\subseteq$ settings.

Def $F_W$ for irreducible complex

Take $x \in C(Y_1)$

$$F_W(x) := \lambda(x) = \sum_{y \in C(Y_2)} \sum_{\mathfrak{s}\ \dim(M_{\mathfrak{s}})=0} \#M_{\mathfrak{s}}(x, W^*, y)\, y$$

Note that unlike for $Y^3 \times \mathbb{R}$, $W$ has no $\mathbb{R}$-action, so we don't quotient by $\mathbb{R}$.

Claim $F_W$ is a chain map: $F_W \circ \partial_{Y_1} = \partial_{Y_2} \circ F_W$.

Pf Exercise

The more interesting definitions are those for the upper and lower bar complexes. Recall that $\bar{C}$ is formed from the irred. complex by incorporating flows into the reducible. We take $C$ and add a copy of the field we're working in at grading 0, viewing it as the generator corresponding to $\Theta$: $\bar{C} = C \oplus \mathbb{F}_{[0]}$. The differential $\bar{\partial} = \partial + D_1$, and its convenient to view $\bar{\partial}$ as having components, eg. $\bar{\partial} = \begin{bmatrix} \partial & 0 \\ D_1 & 0 \end{bmatrix}$. The upper right

entry is 0, since by definition of $\bar{C}$, there are no maps from reducible components to irreducibles. The lower right entry is zero since the differential of the reducible can't be supported on the reducible, since the differential drops the grading by one.

To define $F_W$ in the $\bar{C}$-context, we'll also define it on components. For simplicity, assume that $H_1(W;\mathbb{Z}) = 0$.

Then $F_W : \bar{C}(Y_1) \to \bar{C}(Y_2)$ is given by $F_W = \begin{bmatrix} \lambda & 0 \\ \Delta & 1 \end{bmatrix}$, where

$\lambda$: Irreducible generators $\longrightarrow$ Irreducible generators

$\Delta_1$: Irreducible generators $\longrightarrow$ Reducible generators

$1$: Reducible generators $\longrightarrow$ Reducible generators

$0$: Reducible generators $\longrightarrow$ Irreducible generators

This motivates the following formal definition.

**Def** $F_W$ for $\bar{C}$

When $H_1(W,\mathbb{Z}) = 0$,

$$F_W(x) = \begin{bmatrix} \lambda & 0 \\ \Delta & 1 \end{bmatrix}(x), \quad \text{where} \quad x \in C \oplus \mathbb{F}_{(0)},$$

$$\lambda(x) = \sum_{y \in C(Y_2)} \sum_{\substack{r \\ \dim(\mathcal{M}_r)=0}} \#\mathcal{M}_r(x, W^*, y)\, y \qquad \text{as before,}$$

$$\Delta_1(x) = \sum_{\substack{r \\ \dim(\mathcal{M}_r)=0}} \#\mathcal{M}_r(x, W^*, \Theta)\, Q,$$

and the one in the lower right represents the obvious trivial connection on $W^*$.

In general, when $H_1(W,\mathbb{Z}) \neq 0$, $F_W$ is defined to have components

$$F_W = \begin{bmatrix} \lambda & 0 \\ \Delta_1 & c \end{bmatrix} \quad \text{where} \quad c = |H_1(W,\mathbb{Z})|.$$

The 1 (or c) from Reducibles to Reducibles can be viewed more explicitly. When $H_1=0$, pick the trivial bundle on $W^*$ agreeing with the trivial bundles on each end. The 0 connection on each end and this trivialization defines a 0 connection over all of $W^*$ which clearly sends $\Theta$ to $\Theta$. When $H_1 \neq 0$, this component also roughly has the interpretation as counting ASD connections going from $\Theta$ to $\Theta$, but it turns out that there are $H_1$-many reducible flat connections over $W^*$.

Claim $F_W \circ \overline{\partial} = \overline{\partial} \circ F_W$

Pf Exercise

Rmk The map $\lambda$ can be thought of like the $D_1$ map by replacing $W$ with $Y \times \mathbb{R}$ and quotienting by $\mathbb{R}$.

Of course the $\subseteq$ complex is very similar to that of $\overline{C}$: we disallow flows into $\Theta$, but allow those out of $\Theta$.

Def $F_W$ for $\subseteq$

$$\subseteq = C \oplus \mathbb{F}(-3)$$

$$\underline{\partial} = \partial + D_2 = \begin{pmatrix} \partial & D_2 \\ 0 & 0 \end{pmatrix}$$

$$F_W = \begin{pmatrix} \lambda & \Delta_2 \\ 0 & c \end{pmatrix}$$

where $\lambda, c$ are as before and

$$\Delta_2[\theta] = \sum_{y \in C(Y_2)} \sum_{\substack{x \\ \dim=0}} \#\mathcal{M}_x(\theta, W^*, y)\, y$$

We now ask: What does this formalism buy us?

# Obstructing Cobordisms

Let's work with the underbar complex, $\underline{C}$.

We can also think about the $D_2$ map as giving us the info of a Package Class of the irred. complex $C(Y)$.

We'll call $D_2(\Theta)$ the "$D_2$-class", eg, $[D_2^Y] := [D_2(0)] = [D_2(1)]$.

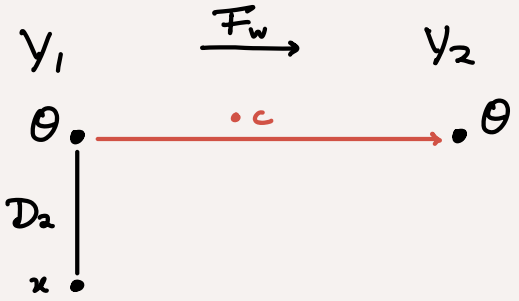
The (rather boring) observation is that a neg.-def. $b_1 = 0$ cobordism $W$ scales the $D_2$ class by $c$.

**Thm** Let $W$ be a negative definite $b_1 = 0$ cob. from $Y_1$ to $Y_2$. Set $c = |H_1(W, \mathbb{Z})|$. Then $F_W[D_2^{Y_1}] = c[D_2^{Y_2}]$

**Pf** Consider the following complexes for $Y_1$, $Y_2$.



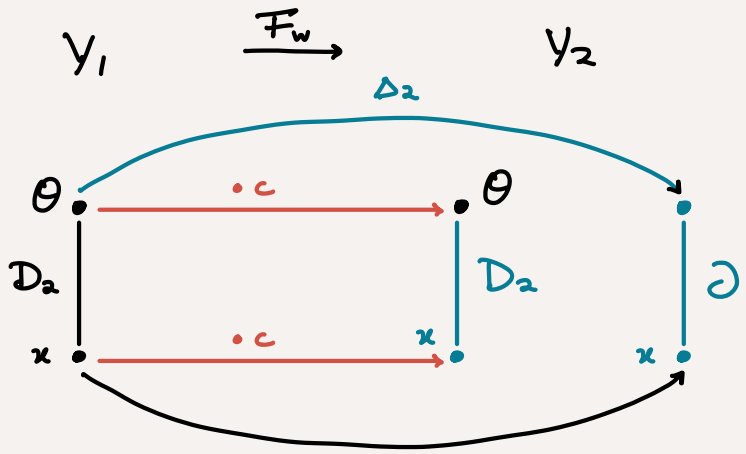

Since reducibles must map to reducibles, there must be an arrow like so




Note that this is related to the question from last time: why define $\underline{C}$ and $\overline{C}$? It's these definitions that give us the map $c$ from red. to red., so we know there

is an arrow.

There might be other arrows and other generators, but we've asserted that $F_w$ is a chain map, so $F_w(x)$ must also multiply by $c$.




Formally, $(F_w \circ \partial)(\theta) = (\partial \circ F_w)(\theta)$

$$F_w(D_2(\theta)) = (\partial \circ F_w)(\theta)$$

$$= (\partial + D_2)(F_w(\theta))$$

$$= (\partial + D_2)(c\theta + \Delta_2\theta)$$

$$= \partial(c\theta) + D_2(c\theta) + \partial(\Delta_2\theta) + D_2(\Delta_2\theta)$$

$$= cD_2\theta + \partial(\Delta_2\theta)$$

Since $\partial(c\theta) = 0$ and $D_2(\Delta_2\theta) = 0$.

($c$ is the constant map, so $D_2(c\theta) = cD_2\theta$) $\square$

Note that its instructive to unite at what $\partial \circ F_w = F_w \circ \partial$ actually means:

$$\begin{pmatrix} \partial & D_2 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} \lambda & \Delta_2 \\ 0 & C \end{pmatrix} = \begin{pmatrix} \lambda & \Delta_2 \\ 0 & C \end{pmatrix}\begin{pmatrix} \partial & D_2 \\ 0 & 0 \end{pmatrix}$$

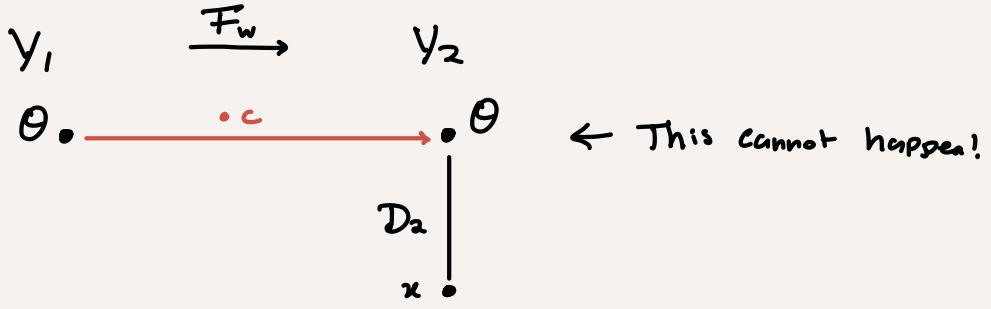
$$\begin{bmatrix} \partial\lambda & \partial\Delta_2 + D_2 C \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \lambda\partial & \lambda D_2 \\ 0 & 0 \end{bmatrix}$$

So this equality implies a more CPX relationship btwn the maps $\partial, D_2, \lambda, \Delta_2, C$.

As a consequence, can obstruct cobs. by calculating $D_2$ maps.

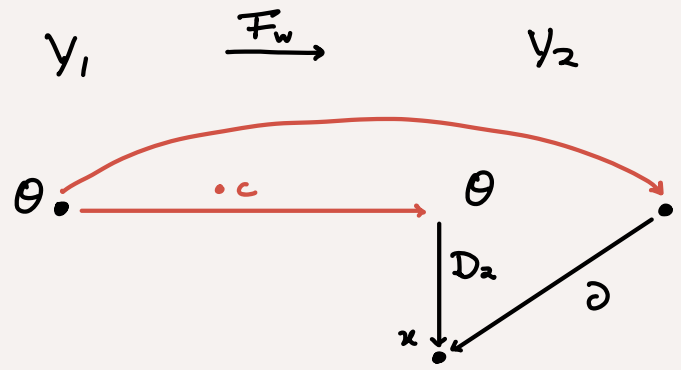
**Cor** If $[D_2^{Y_1}] = 0$ but $[D_2^{Y_2}] \neq 0$, then $\nexists$ neg-def $b_1 = 0$ cob. W from $Y_1$ to $Y_2$

Intuitively, cobordism maps take "red to red", ie, $\Theta \mapsto C\Theta +$ (other stuff).
Therefore, there can be no situation where $\Theta$ maps to nothing but $F_W(\Theta)$ does map to something:




← This cannot happen!

However, the claim is not that $D_2 = 0$, but $[D_2] = 0$, so it is o.k. for the complex to look like this:




This is the ethos of Floer theory: Say some cob. map has some formal prop, then show that $\exists$ any map w/ that prop.

Rmk 1) This is analog of the d-inv being a particular sign (there is a more precise statement of this)

2) $D_1$ gives a coh. class. Doles of $Y_1, Y_2$ will switch

3) Just like for cl. ft, all cob maps lift to $\mathbb{Z}$-version

So far, this discussion is v. similar to other FHxs. $|FH|$ is special because it admits a filtration by CS value; we have not used that yet in this discussion

## Cobordism maps & the CS Filtration

The overarching claim here is that everything we've done so far is actually filtered.

Thm $F_W$ is filtered.

Pf If $y$ appears in $F_W(x)$, then there is some ASD conn. $A$ limiting to $x$ and $y$ on the ends. This can be thought of as "$A$ restricts to $x$ and $y$ on each boundary"; but this is slightly false; the analysis prefers open boundaries

to fixed conditions on closed boundaries.

Then, again playing fast and loose with orientations, we get that

$$\widetilde{CS}(y) - \widetilde{CS}(x) = \frac{1}{8\pi^2}\int_W \mathrm{tr}(F_A \wedge F_A)$$

$$= \frac{-1}{8\pi^2}\int_W \mathrm{tr}(F_A \wedge *F_A)$$

$$= -\|F_A\|^2_{L^2} \leq 0$$

Thus $\widetilde{CS}(y) \leq \widetilde{CS}(x)$. □

So $F_W$ is filtered exactly because the CS functional is a norm.

**Thm** Spse $x, y$ corresp. to $SU(2)$-reps of $\pi_1(Y_1)$ and $\pi_1(Y_2)$. If $y$ appears in $F_W(x)$ and $\widetilde{CS}(x) = \widetilde{CS}(y)$, then $\exists$ $SU(2)$ rep$^n$ of $\pi_1(W)$ which restricts to that of $x$ and $y$. Eg, given representations $\rho_1 : \pi_1(Y_1) \to SU(2)$ and $\rho_2 : \pi_1(Y_2) \to SU(2)$ there is a representations $\rho : \pi_1(W) \to SU(2)$ st $\rho|_{\pi_1(Y_i)} = \rho_i$.

**Pf** If $\widetilde{CS}(x) = \widetilde{CS}(y)$, $-\|F_A\|^2_{L^2} = 0$, so $A$ has no curvature, and defines a rep$^n$ of $\pi_1(W)$. □

The obvious takeaway is that the cobordism map on $|\mathbb{F}|$ is affected by $\pi_1$ of the cob. But what this is actually saying is subtle: the very existence of cobs is determined by rep's of $\pi_1$. This gives the following corollary.

**Cor** If $\pi_1(W) = 1$, then $F_W$ is strictly filtered except for the $\Theta$-to-$\Theta$ component.

Again, so what? We actually have enough formal props to conclude our first application! Recall that on the first day, we asked whether the Poincaré hs sphere $\Sigma(2,3,5)$ admits a simply connected self-homology cob. We claimed that this was an interesting question, since it's equivalent to asking whether $\Sigma(2,3,5) \# -\Sigma(2,3,5)$ bounds a simply connected homology ball. The underlying question is: Does every $\mathbb{Z}HS^3$ bound a contractible $\mathbb{Z}HB^4$?

The point here is that instantons can detect this, in this subtle way. Lets look at the IFH of the Poincaré sphere. This does, unfortunately, open the can of worms which asks how does one compute this? But we'll punt this question to later.

Example IF complex of $\Sigma(2,3,5)$

$gr(\theta) = -3$ $\theta \bullet$ $CS(\theta) = 0$

$\downarrow D_2$

$gr(\alpha) = -4$ $\alpha \bullet$ $CS(\alpha) = \frac{-1}{120}$

$gr(\beta) = -8$ $\beta \bullet$ $CS(\beta) = \frac{-49}{120}$

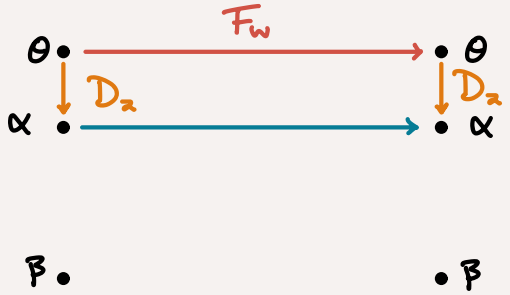
Is it possible that there is a $\pi_1 = 1$ self-hs cob?

If so, it induces a cob map.

Since $H_1 = 0$, $\theta \mapsto \theta$, and since

$F_W$ commutes with $D_2$, $\alpha \mapsto \alpha$.

However, sending $\alpha$ to itself is not strictly filtered.



Therefore $W$ cannot be simply connected.

It is important here that we knew smth abt the $D_2$ map. If we didnt know abut $\Theta$, theres no reason why $F_W$ cant be 0. But $\Theta$ tells us that $\Theta \mapsto \Theta$, so this "shadow" of the $D_2$ map forces $\alpha \mapsto \alpha$. This is actually a completely general fact.

**Thm** If $[D_2{}^Y] \neq 0$, then there does not exist any simply connected homology cobordism from $Y$ to $Y$

**Pf** Choose a rep. of $[D_2{}^Y]$ in $H_*(C(Y))$ with minimal CS filtration. (The filtr. of the chain cpx induces filtr. on $H_*$). This element can be written as $\sum_i c_i x_i \in H_*(C(Y))$, and $\text{filt}\left(\sum_i c_i x_i\right) = \max \text{filt}(x_i)$ so

$$\text{filt}([x]) = \min \text{filt. of rep}^n\text{'s of } [x]$$

If there was a simply connected homology cobordism $W: Y \to Y$, then $\text{filt}(F_W(x)) < \text{filt}(x)$. But $F_W(x)$ is also a rep. of $[D_2{}^Y]$, implying our og. choice for $x$ was not minimal. $\Rightarrow\Leftarrow$ $\square$

So "most" things dont admit s.c. self Hx. cobs.